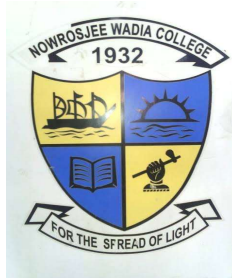


Modern Education Society's
Nowrosjee Wadia College, Pune (Autonomous)



Department of Computer Science
T.Y.B.Sc (Information Technology) (NEP 2.0)

MINI PROJECT REPORT
ON
“STUDENT PERFORMANCE PREDICTION”
A Supervised Machine Learning Project Using Student Performance Data

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1.Introduction

Student performance prediction is an important application of data analysis and machine learning in the field of education. Student academic performance can be influenced by several factors such as study time, attendance, previous scores, family background, and other personal and academic factors. Analyzing these factors can help identify patterns in student performance.

In this project, a student performance dataset is analyzed using statistical analysis and machine learning techniques. The dataset is first explored and preprocessed to understand the different variables and their relationships. Exploratory Data Analysis (EDA) is performed using suitable statistical measures and visualizations. A supervised machine learning model is then trained to predict student performance based on the available input features.

The main purpose of this project is to understand how machine learning can be applied to educational data and how the resulting model can be used for student performance prediction.

2. Problem Statement

Student performance is affected by various academic, personal, and environmental factors. It can be difficult to identify the factors that have the greatest influence on a student's academic results by using traditional methods alone. Therefore, there is a need for a data-driven approach that can analyze student-related information and predict academic performance.

The problem addressed in this project is to develop a supervised machine learning model that can use the available student data to predict student performance. The project involves analyzing the dataset, preprocessing the data, identifying important patterns through exploratory data analysis, and applying a suitable machine learning algorithm for prediction.

3. Objectives

- 1.To analyze the student performance dataset and understand its different attributes.
- 2.To perform data preprocessing and prepare the dataset for machine learning.
- 3.To identify patterns and relationships between different factors affecting student performance.
- 4.To apply a suitable supervised machine learning algorithm for predicting student performance.
- 5.To evaluate the performance of the trained machine learning model using appropriate evaluation measures.
- 6.To understand the practical application of machine learning in the field of education.
- 7.To identify the factors that influence student academic performance.
- 8.To apply suitable machine learning algorithms for student performance prediction.
- 9.To evaluate the performance of the developed machine learning model.
- 10.To use the trained model to predict student performance based on input features.

4. Dataset Description

The dataset used in this project contains information about students and various academic, personal, family, and social factors that may be related to student performance. The dataset contains 395 student records and 33 attributes. The data includes both categorical and numerical variables. The final grade attributes G1, G2, and G3 represent student grades and are used for analyzing academic performance.

The dataset contains both numerical and categorical attributes. Categorical attributes are converted into numerical form using one-hot encoding before applying the machine learning models.

DATASET DETAIL	DESCRIPTION
DATASET NAME	Student Performance Dataset
FILE USED	Student-mat.csv
NUMBER OF RECORDS	395
NUMBER OF COLUMNS	33
TARGET	Pass/Fail
PROGRAMMING LANGUAGE	Python
LEARNING METHOD	Supervised Learning

5 . TOOLS AND TECHNOLOGIES

Programming Language: Python

Development Environment: Jupyter Notebook

Libraries Used:

Pandas — for data handling and analysis

NumPy — for numerical operations

Matplotlib — for data visualization

Seaborn — for statistical visualization

Scikit-learn — for machine learning and model evaluation

Machine Learning Algorithms:

Logistic Regression

Decision Tree Classifier

Random Forest Classifier

Dataset: Student Performance Dataset (student-mat.csv)

Learning Method: Supervised Learning

6. METHODOLOGY

The methodology followed in this project consists of the following steps:

1. Data Collection: The Student Performance Dataset was obtained from a publicly available source.

2. Data Loading: The student-mat.csv file was loaded into Python using the Pandas library.

3. Data Understanding: The dataset was examined using functions such as `head()`, `info()`, `describe()`, and `isnull().sum()`.

4. Target Creation: A new target variable called `pass` was created. Students with a final grade $G3 \geq 10$ were classified as Pass, while students with $G3 < 10$ were classified as Fail.

5. Data Preprocessing: The final grade `G3` and target variable were removed from the input features. Categorical variables were converted into numerical form using one-hot encoding.

6. Data Splitting: The dataset was divided into 80% training data and 20% testing data.

7. Model Training: Three supervised learning algorithms were applied:

Logistic Regression

Decision Tree

Random Forest

8. Model Evaluation: The models were evaluated using accuracy, classification reports, and a confusion matrix.

9. Prediction: The trained Random Forest model was used to predict the pass/fail result for a sample student.

7. DATA PREPROCESSING

Data preprocessing is an important step in machine learning. It prepares the dataset so that it can be used effectively for model training.

The following preprocessing steps were performed:

1. Handling Missing Values: The dataset was checked for missing values using `isnull().sum()`.

2. Target Variable Creation: The final grade G3 was converted into a binary target variable named pass.

1 = Pass

0 = Fail

3. Removing Unnecessary Columns: The final grade G3 and the newly created pass column were removed from the input features.

4. Encoding Categorical Data: Categorical variables were converted into numerical variables using one-hot encoding.

5. Feature Scaling: StandardScaler was used to scale the features for the Logistic Regression model.

6. Train-Test Split: The dataset was divided into 80% training data and 20% testing data.

This preprocessing ensures that the data is in a suitable format for applying machine learning algorithms.

8.EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis (EDA) was performed to understand the dataset and identify useful patterns in student performance.

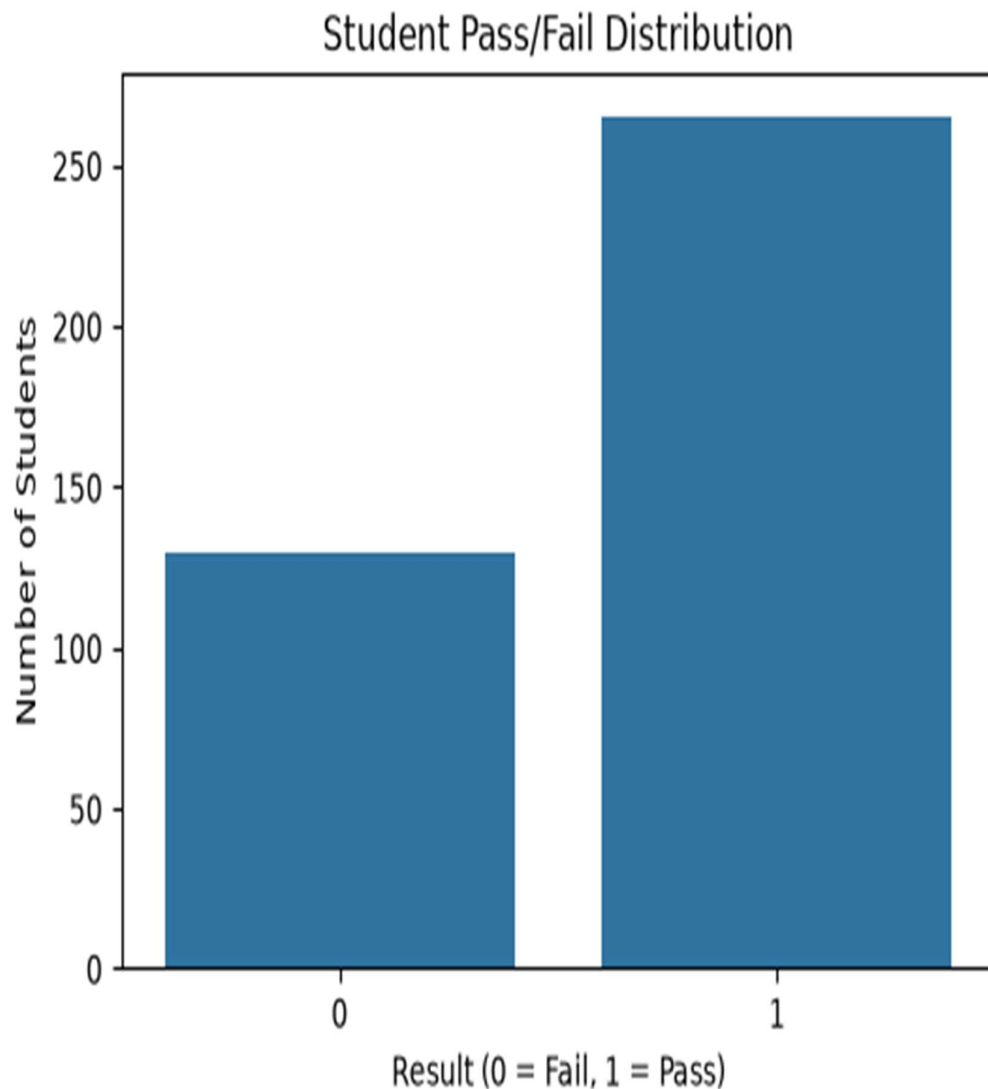
The following analyses were performed:

Pass/Fail Distribution: A count plot was created to show the number of students who passed and failed.

Correlation Analysis: A correlation heatmap was created to understand the relationships between numerical variables in the dataset.

Dataset Summary: Functions such as `head()`, `info()`, and `describe()` were used to understand the structure and statistical properties of Data

The visualizations helped in understanding the distribution of student results and the relationships between different numerical attributes.



Interpretation of Pass/Fail Distribution

> The graph shows the distribution of students according to their final result. Students are divided into two categories: Pass (1) and Fail (0). The graph helps us understand how many students belong to each category. It also shows whether the dataset contains a balanced or unequal distribution of pass and fail students.

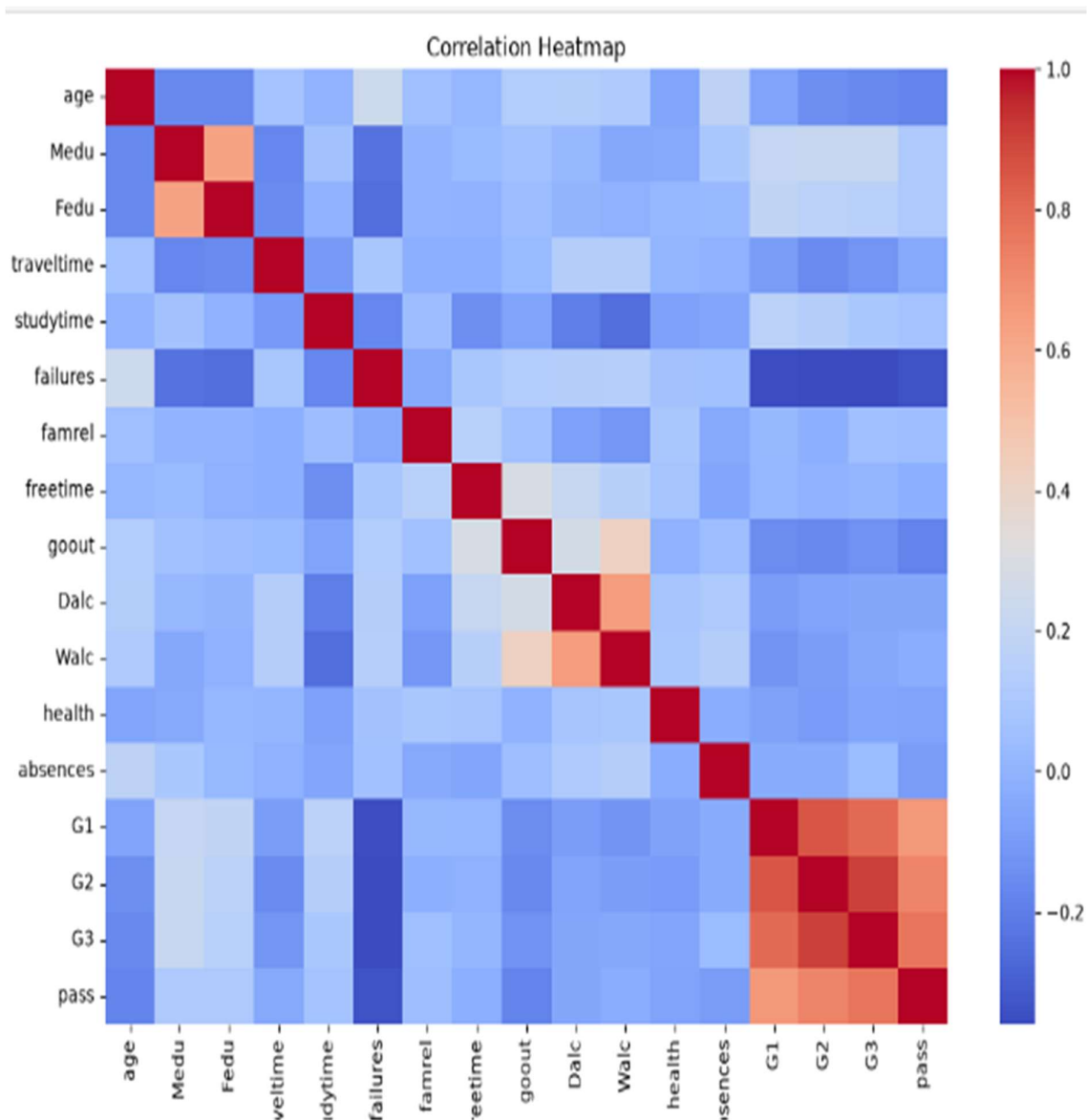
> The Pass/Fail Distribution graph represents the number of students who passed or failed based on their final grade. In this project, the final grade G3 is used to create the target variable. A student with $G3 \geq 10$ is classified as Pass (1), while a student with $G3 < 10$ is classified as Fail (0).

The graph provides a clear visual representation of the distribution of students between the two result categories. It helps us understand how the students in the dataset are divided according to their academic outcome.

The distribution is useful for checking whether the two classes are approximately balanced or whether one class contains considerably more students than the other. This is important because an unequal class distribution can affect the performance and interpretation of a classification model.

The graph also provides an initial understanding of the target variable before applying machine learning algorithms. Based on this target variable, the models are trained to classify a student's result as either Pass or Fail.

Therefore, the Pass/Fail Distribution graph is an important part of the Exploratory Data Analysis because it gives a simple overview of the target classes used in the prediction task.



Interpretation of Correlation Heatmap

> The correlation heatmap shows the relationships between numerical variables in the dataset. The values closer to +1 indicate a strong positive relationship, while values closer to -1 indicate a strong negative relationship. Values near 0 indicate a weak relationship. The heatmap helps identify which numerical features are related to student performance.

A value close to +1 indicates a strong positive relationship, meaning that when one variable increases, the other tends to increase as well. A value close to -1 indicates a strong negative

relationship, meaning that when one variable increases, the other tends to decrease. A value close to 0 indicates little or no linear relationship.

The heatmap helps us identify important relationships between different student-related factors and academic performance. It is useful during Exploratory Data Analysis because it provides a visual overview of the numerical relationships in the dataset.

In this project, the correlation heatmap is mainly used to understand the dataset before applying machine learning models. Correlation does not necessarily mean that one variable causes another.

A value close to +1 indicates a strong positive relationship, meaning that when one variable increases, the other tends to increase as well. A value close to -1 indicates a strong negative relationship, meaning that when one variable increases, the other tends to decrease. A value close to 0 indicates little or no linear relationship.

The heatmap helps us identify important relationships between different student-related factors and academic performance. It is useful during Exploratory Data Analysis because it provides a visual overview of the numerical relationships in the dataset.

9.MACHINE LEARNING MODELS

Three supervised machine learning algorithms were used in this project to predict whether a student would Pass or Fail.

9.1 Logistic Regression

Logistic Regression is a classification algorithm used to predict one of two possible outcomes. In this project, it predicts whether a student will Pass (1) or Fail (0) based on the available student features.

9.2 Decision Tree

A Decision Tree is a classification algorithm that makes decisions using a tree-like structure. It divides the data into different groups based on feature values and finally predicts the student's result as Pass or Fail.

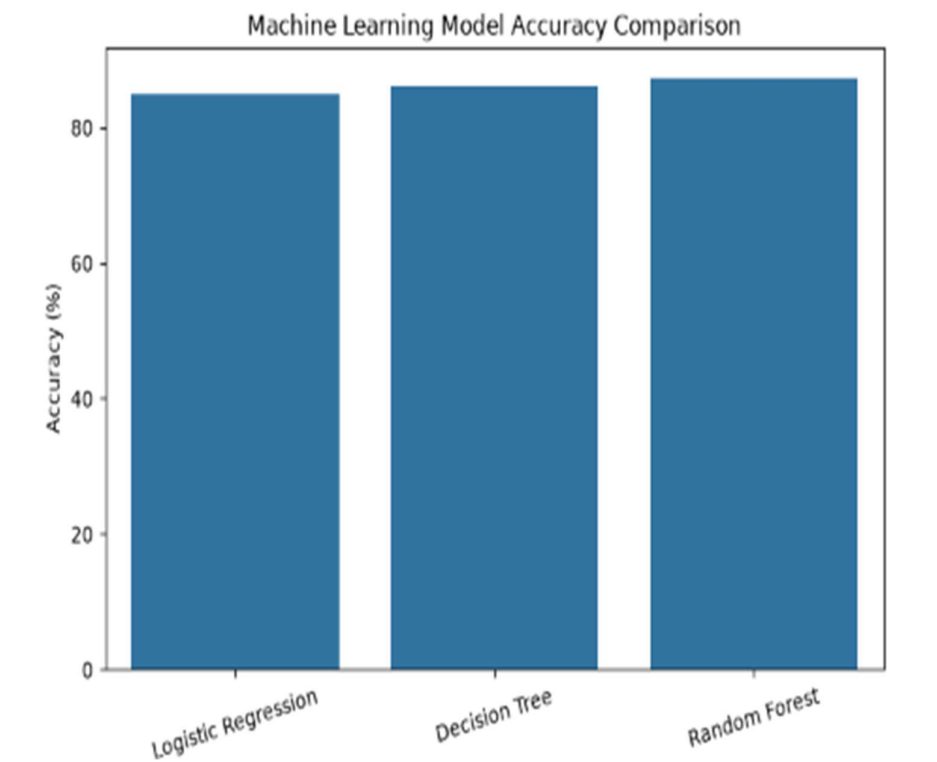
9.3 Random Forest

Random Forest is an ensemble machine learning algorithm that combines multiple Decision Trees to make a prediction. It generally provides more stable predictions by considering the results of several trees.

Model Training

The dataset was divided into 80% training data and 20% testing data. The training data was used to train the models, while the testing data was used to evaluate their performance.

10.Accuracy Comparison of Machine Learning Models



Interpretation of Accuracy Comparison

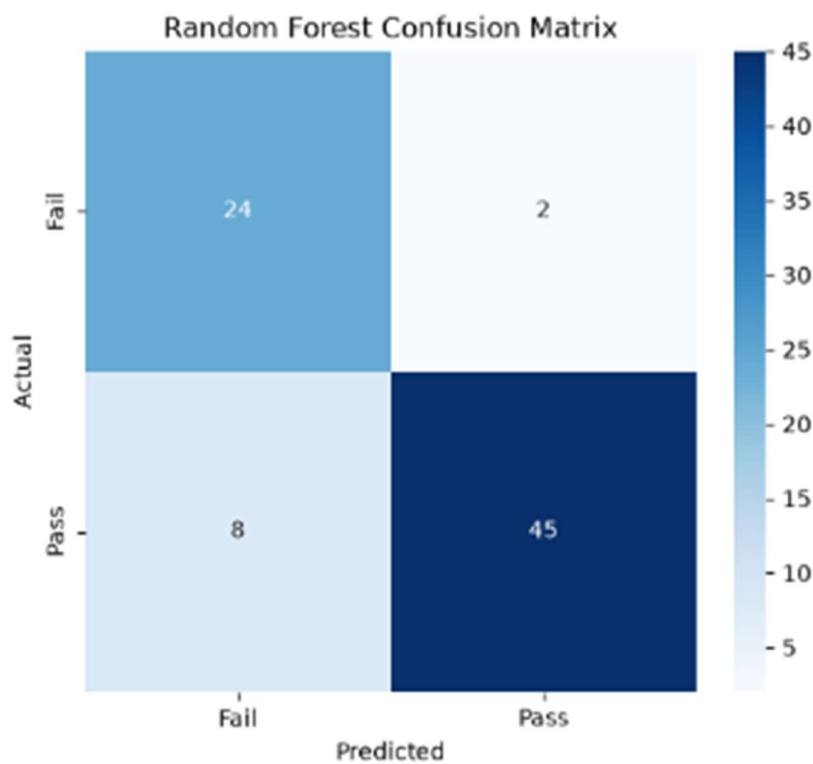
> The accuracy comparison graph shows the performance of the three machine learning models used in this project. Logistic Regression achieved an accuracy of 84.81%, Decision Tree achieved 86.08%, and Random Forest achieved 87.34% on the testing dataset.

The three models achieved different levels of accuracy when applied to the same dataset. The graph makes it easier to visually compare the accuracy obtained by each model. The results show that.

Accuracy is calculated based on the number of correctly classified test samples compared with the total number of test samples. The accuracy results are used to understand the classification performance of the models on the testing data.

11. Confusion Matrix of Random Forest Model

The confusion matrix provides a detailed view of the correct and incorrect classifications made by the Random Forest model on the testing dataset. It helps in understanding the model's classification performance beyond the overall accuracy.



12. SAMPLE STUDENT PREDICTION

The trained Random Forest model was used to make a prediction for a sample student. The model uses the student's available features to predict whether the student is likely to Pass or Fail.

In the sample prediction performed in Python, the model predicted:

Predicted Result: PASS

This demonstrates how the trained machine learning model can be used to classify a student's expected academic outcome.

13. CLASSIFICATION REPORT

> The classification report provides detailed information about the performance of the machine learning model. It includes precision, recall, F1-score, and support for each class.

Precision shows how accurately the model identifies a particular class. Recall shows how many of the actual samples of a class were correctly identified. F1-score combines precision and recall into a single measure. Support represents the number of actual samples belonging to each class in the testing dataset.

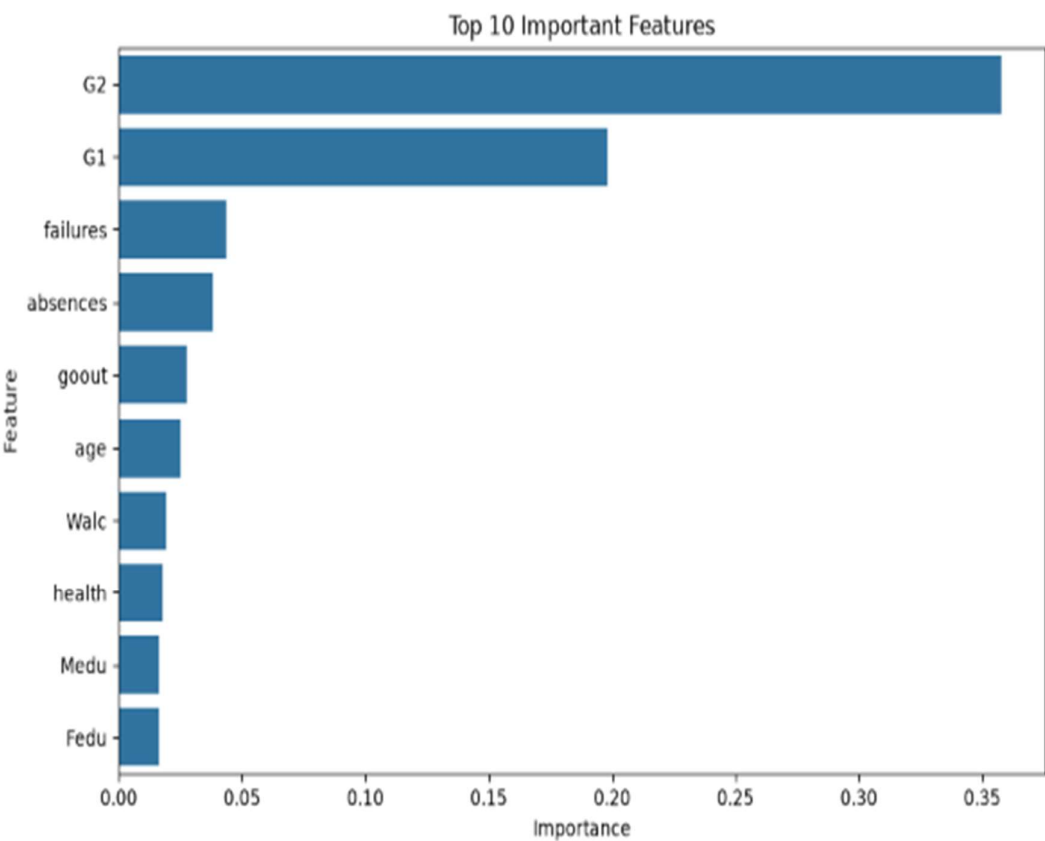
In this project, the classification report was generated for the Random Forest model to understand its Pass/Fail classification performance in greater detail.

14. FEATURE IMPORTANCE

> Feature importance shows which input features contributed most to the predictions made by the Random Forest model. It helps in understanding which variables had a greater influence on the model's classification decisions.

The Random Forest algorithm calculates an importance value for each feature. Features with higher importance values contributed more to the model's predictions, while features with lower values had less influence.

The feature importance graph provides a visual comparison of the contribution of different student-related attributes. This helps in interpreting the machine learning model and understanding the factors considered important during prediction.



15. MODEL EVALUATION AND RESULTS

The trained machine learning models were evaluated using accuracy. Accuracy measures the percentage of test samples that were classified correctly.

The obtained results are:

Machine Learning Model	Accuracy
Logistic Regression	84.81%
Decision Tree	86.08%
Random Forest	87.34%

Result Analysis

The three models produced different accuracy values on the testing data. Logistic Regression achieved an accuracy of 84.81%, while Decision Tree achieved 86.08%. Random Forest achieved 87.34%.

These results show how different supervised learning algorithms performed on the same student dataset. The accuracy values are based on the 20% testing portion of the dataset.

Important: Accuracy is only one evaluation metric; it does not by itself describe every aspect of model performance.

14. CONCLUSION

> This project successfully demonstrated the use of supervised machine learning for predicting student academic outcomes. The Student Performance Dataset was analyzed using Python, followed by data preprocessing and exploratory data analysis.

Three machine learning algorithms—Logistic Regression, Decision Tree, and Random Forest—were implemented and evaluated. Their performance was compared using accuracy on the testing dataset.

The project shows that machine learning can be used to analyze student-related data and classify students into Pass or Fail categories. It also provided practical experience in data preprocessing, visualization, model training, and evaluation.

15. FUTURE SCOPE

> The project can be further improved by using a larger and more diverse student dataset. Additional student-related factors can also be included to improve the analysis.

In the future, other machine learning algorithms can be tested and their performance can be compared using multiple evaluation metrics.

The project can also be developed into a simple web-based application where teachers or students can enter relevant information and receive a predicted Pass/Fail result.

Further improvements could include regular model retraining with new student data and better visualization of student performance patterns.

REFERENCES

Student Performance Dataset — UCI Machine Learning Repository.

Python Documentation — Python programming language documentation.

Pandas Documentation — Data analysis and manipulation library.

Scikit-learn Documentation — Machine learning library for Python.

Matplotlib Documentation — Data visualization library.

Seaborn Documentation — Statistical data visualization library.